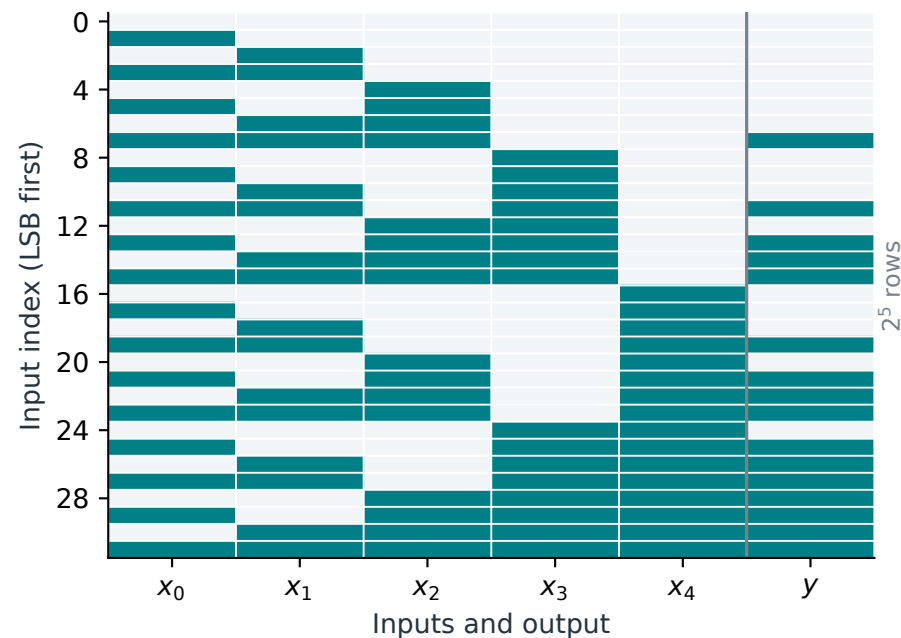


Index deconvolution: from exact behaviour to a checked generative description

A. Exact repertoire supplied to the inverse



B. Perturbation recovers functional inputs

Compare $y[x]$ with $y[x \oplus 2^i]$ for each i .

bit	x	$x \oplus 2^i$	y	y' (change)
0	6	7	0	1
1	5	7	0	1
2	3	7	0	1
3	3	11	0	1
4	3	19	0	1

$l_c =$ **connected inputs** = **essential variables**

Here every coordinate has a witnessed output change.

C. Lossless schema compression

Output-one states are grouped by fixed bits and don't-care positions.

111** 1**11
 11*1* *111*
 11**1 *11*1
 1*11* *1*11
 1*1*1 **111

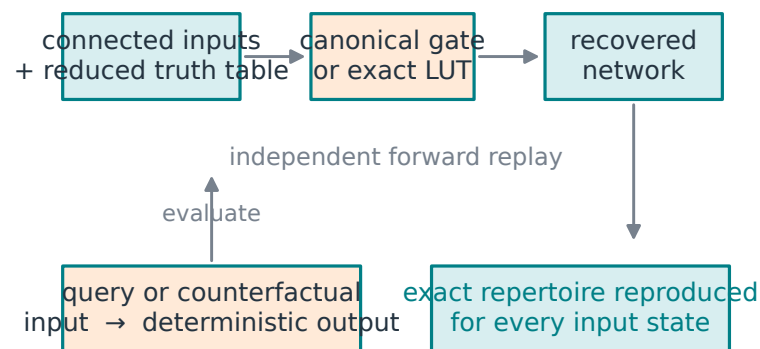
111** : $P = 1 + 2 + 4 = 7$

$S = \{0, 8, 16, 24\}$

$\text{Dec}(P, S) = \{7, 15, 23, 31\}$

schema union = exact on-set
(16 output-one indices)

D. Reconstruct, replay, and answer queries



The guarantee is exact within the stated model and data domain; it is not a claim of minimal K or unique hidden wiring.